

```
$ mysql -e "SHOW CREATE TABLE orders" --normalize
```



// 1NF → 2NF → 3NF □□□□□

1NF

□□□

2NF

□□□□□□

3NF

□□□□□□□

□□□□

OLAP □□□

\$ cat schema-plan.md



01 / 

□□□□□□□□

02 / 

1NF    → 2NF → 3NF

03 / 

JOIN □□□

04 / 

OLTP vs OLAP □□□

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PART 01



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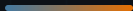
order_id	customer_name	customer_email	product_name	price
1	□□	<u>zhang@example.com</u>	□□□□□	30000
2	□□	<u>zhang@example.com</u>	□□	500
3	□□	<u>li@example.com</u>	□□	1200

□□□□

□□□□□□□□ email □□□□

□□□□

□□□ email□□ UPDATE □□□□



ROOT CAUSE



CHECKPOINT_01

データベースの UPDATE 37 回実行後 3 分経過

A データが更新された

B データが更新されなかった

C データが更新された

PART 02



1NF → 2NF → 3NF → ☐ ☐ ☐ ☐ ☐ ☐

1NF

NO DUPLICATES

BEFORE — 1NF

student	courses
11	Maths, Maths, Maths, Maths, Maths, Maths, Maths

AFTER — 1NF

student	course
11	Maths
11	Maths
11	Maths

1NF

2NF

3NF

2NF NOGRAPH

BEFORE — □□□□

order_id	product_id	product_name	qty
1	P1	□□	1
2	P1	□□	3

AFTER — □□□□□

order_id	product_id	qty
1	P1	1
product_id	product_name	
P1	□□	

3NF

NO SLIP

□□□□□□

student_id → dept_id → dept_name — dept_name — dept_id —

BEFORE — □□□□

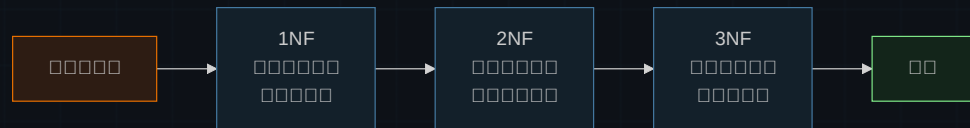
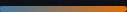
student_id	name	dept_id	dept_name
S1	□□	D1	□□□
S2	□□	D1	□□□

AFTER — □□□□□

student_id	name	dept_id
S1	□□	D1
dept_id	dept_name	
D1	□□□	

□□□□□□□□□□ — □□□□□□□□□□

□ □ □ □ □ □ □



3NF 3 columns 6 columns 4 rows

PART 03



□□□□□□JOIN □□□

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JOIN □□
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JOIN □□□□□

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OLTP vs OLAP

OLTP	OLAP	Comparison
Transactional	Analytical	
Real-time	Batch	
High write volume	High read volume	
Simple queries	Complex queries	
Small result sets	Large result sets	
OLTP	OLAP	

STRATEGY

OLTP is designed for high transaction volume and low latency. OLAP is designed for high query volume and high latency.



→



1NF     → 2NF     → 3NF    

3

NF

1

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OLTP □□□□OLAP □□□□□□□□

Schema normalized.

XXXXXXXXXXXX JOIN XXXXX

```
$ mysql -e "EXPLAIN SELECT ..."
```

1NF → 2NF → 3NF → XXXXXXXX